

THE GENERALIZED EXISTENCE THEOREM

A Unified Formal Framework for the M/I/C Trinitarian Minimum and Its Foundational Implications

PAUL BRYANT VANSLYKE

Natural Organization Institute | Lyons, Colorado | 2026

Developed through live dialectical stress-testing against multiple AI reasoning systems. Every escape route examined. Every seam closed.

Abstract

We formalize the functional requirements for persistent systems ($X_f > 0$) within an entropic environment. Using the Generalized Existence Equation (GEE) as an organizing index, we establish that Meaning (Mmin), Intelligence (Imin), and Loop Closure (Cmin) are inseparable, foundational minima required for persistence. The removal of any component leads to immediate systemic dissipation, establishing $M:I:C = 1$ as a primary logical necessity.

Because these minima are foundational rather than emergent, all derived systems — including nucleotide chains, coding structures, and complex life — are constrained by intelligence. The Causal Continuity Axiom demonstrates that functional transitions observed at every stage of biological history presuppose $\Phi > 0$ continuously, not episodically. The Axiom is supported by five independent derivations and is presented as a derived conclusion rather than an assumed premise.

Version 7 adds Section 1.4: The Determination Anchor — formalizing the Anchor-Single pair analysis (61,320,087 pairs; 1,527,212 functional sequences) as direct empirical confirmation that the system is non-random at the operational floor. Non-random equals determined by definition. Determined equals intelligent by functional derivation — through a four-step logical chain establishing that determination requires a comparator, a comparator requires discernment, discernment cannot be blind, and that functional description is what intelligence means stripped of metaphor.

Version 7 further adds the Darwin Falsification Statement: Darwin specified that his theory would absolutely break down if any complex organ or system were found that could not be formed by numerous successive slight modifications. Intelligence — established as the capacity to set boundaries between what occurs and what does not — is not produced incrementally from non-intelligence. Zero boundary-setting capacity produces zero boundary-setting. This satisfies Darwin's own stated falsification condition on Darwin's own terms.

UNIFYING SENTENCE: The GEE indexes the operational signature of Φ within derived systems. Part IV formalizes the measurable thresholds (T_i , T_p , T_r) that confirm when Φ is genuinely non-zero. The Causal Continuity Axiom establishes

why Φ could never have been zero at the foundation. Section 1.4 anchors the determination argument to 61 million empirically measured Anchor-Single pair observations. The Darwin Falsification Statement closes the argument on Darwin's own terms.

Part I: The Generalized Existence Equation

1.1 The Functional Minima of Persistence

To distinguish persistent phenomena from transient stochastic artifacts, we define the foundational components of the Coordination Protocol (Φ) at their lowest operational threshold:

Component	Name	Engineering Role	Biological Instantiation
Mmin	Meaning / Valence	Reference Signal — The Goal. Without this, 'Function' has no coordinate.	DNA as the stored target state; the sequence that defines what a functional protein is
Imin	Intelligence / Correction	Comparator/Actuator — Error Correction. Identifies deviation and applies corrective ΔI .	Proofreading enzymes; negative feedback loops in gene regulation
Cmin	Loop Closure / Feedback Integration	Feedback Integration — The Wiring. Unifies M and I into a single self-referential persistent entity.	The ribosome as translation protocol: the physical interface connecting digital code (DNA) to analog work (proteins)

These are not smuggled definitions. They are the Minimum Functional Requirements (MFR) for any closed-loop control system in cybernetics. A control system without a reference signal has no error to measure. Without a comparator, it cannot act. Without feedback wiring, M and I are physically isolated facts — not a persistent system.

1.2 The Generalized Existence Equation (GEE)

The GEE is presented as an organizing index — a formal accounting structure that maps the variables governing persistence, not a theorem derived from first principles:

$$Xf(t) = \int [t_0 \text{ to } t_n] (\Phi \cdot \Delta I_{sys} / \Omega_{sys}) - \epsilon_{sys} dt$$

Variable	Definition
ΔI_{sys}	Informational Flux — Total bits of change entering the system

Variable	Definition
Ω_{sys}	Search Space — Total number of possible states (for a single functional protein: $\Omega \approx 10^{70}$)
ϵ_{sys}	Entropy Tax — Constant decay of functional information over time (Second Law)
Φ	Coordination Ratio — Efficiency of the protocol in directing ΔI toward X_{target} . $\Phi = I_{\text{functional}} / I_{\text{total}} \in [0, 1]$

The Φ Definition (Operationalized): $\Phi = I_{\text{functional}} / I_{\text{total}}$. This ratio operationalizes the degree to which informational flux is directed toward functional outcomes rather than distributed stochastically. In a purely stochastic process, $\Phi \rightarrow 0$ and the system is dominated by $-\epsilon_{\text{sys}}$, trending toward dissolution.

1.3 Empirical Grounding of GEE Variables

Prior versions presented GEE variables as conceptually grounded in external literature but numerically unanchored. Version 5 provided biological grounding. Version 7 adds the Anchor-Single determination data as an additional empirical layer. Version 8 (this version) adds the convergence analysis establishing that three independently motivated measurements — cohort recall enhancement (R), CLR-corrected channel independence (C), and directionality deviation (D) — map onto the GEE's M_{min} , I_{min} , and C_{min} structure with predicted relationships, confirming $\Phi > 0$ continuously and the vessel principle across the full sensitivity range of absolute scaling (Van Slyke, 2026g). See Appendix A.

EMPIRICAL ANCHOR: $\Phi_{\text{CI}} = 0.2080$ derived as supremum of observed constraint density values: $\Delta C(N) = 1 - H_{\text{observed}}(N) / H_{\text{max}}(N)$. 16/16 biological measurements show $\Delta C > 0$. Structure is never absent at any complexity level in any organism measured. The gap $G(N) = \Phi_{\text{CI}} - \Delta C(N)$ is non-zero at 15 of 16 measurement points. The source always exceeds the expression (Van Slyke, 2026b).

1.4 The Determination Anchor — The Functional Derivation

This section formalizes the Anchor-Single pair analysis as direct empirical confirmation that the system operates non-randomly at the most local measurable unit of sequence organization — the 5-residue non-overlapping window. It further presents the formal derivation establishing that determination and intelligence are functionally equivalent — not by dictionary definition but by logical necessity derived from the requirements of determination itself.

The Measurement

For every 5-residue non-overlapping window in 1,527,212 functional protein domain sequences (Pfam curated), the dominant Anchor amino acid (appearing ≥ 2 times) and all Single amino acids (appearing exactly once, distinct from the Anchor) were identified. Pair frequencies were compared against expected frequencies under statistical independence using standardized residuals (Z-scores). Bonferroni correction applied for 400 tests.

NEW EMPIRICAL ANCHOR: 61,320,087 Anchor-Single pairs analyzed. 1,527,212 functional sequences (Pfam). 19,630,673 disordered pairs (DisProt, 198,215 sequences). Top admitted pair A $\rightarrow$ L: Z = 239.55, Obs/Exp = 1.3364. Top avoided pair A $\rightarrow$ N: functional Z = -87.13, disordered Z = +2.33 (delta = 89 Z-score points). 89% of attractor-anchored pairs significant after Bonferroni correction. Latent potential: 0.727 functional vs 0.532 disordered. Anti-sequential bonds: 25 functional vs 14 disordered (Van Slyke, 2026d).

The Four-Step Logical Closure

The Anchor-Single data closes the argument in four steps. Steps 1 and 2 require no premises beyond the data and standard definitions. Steps 3 and 4 present the functional derivation that replaces all prior dictionary-based formulations of this step.

Step 1 — The data shows non-randomness at the operational floor. A $\rightarrow$ N avoided at Z = -87.13 in functional sequences while favored at Z = +2.33 in disordered sequences. This is not a continuum. It is a categorical organizational reversal across 61 million observations.

Step 2 — Non-random equals determined by definition. The opposite of random is determined. This requires no additional premise.

Step 3 — Determined equals intelligent by functional derivation. This step is the critical junction and is stated as a derived conclusion through four sub-steps.

Sub-step 3A: Determination requires a comparator. A determined system is one in which prior conditions constrain the probability distribution of subsequent states. For a prior condition to constrain a probability distribution, it must distinguish between possible outcomes. A condition that makes no distinction between states constrains nothing. Therefore determination requires a prior condition that performs differential evaluation of states — a comparator by functional definition.

Sub-step 3B: A comparator requires discernment. A comparator is not a passive physical state merely affected by inputs. To compare is to evaluate states against a criterion and produce a differential response. A system that responds identically to all inputs regardless of their character performs no comparison. Therefore a comparator requires discernment — the capacity to evaluate states differentially against a criterion.

Sub-step 3C: Discernment cannot be blind. A blind process has no criterion against which states are evaluated. A blind process cannot perform differential evaluation because differential evaluation requires a criterion that distinguishes preferred from non-

preferred states. Therefore discernment requires a criterion maintained consistently across instances.

Sub-step 3D: The functional description of what discernment requires is identical to the functional description of intelligence. A system that maintains a criterion, evaluates states against that criterion, and differentially constrains outcomes on the basis of that evaluation is — at its most basic mechanical level, stripped of all metaphor and anthropomorphic loading — what intelligence means functionally. This is not a definition imported from outside the argument. It is the description that falls out of the requirements of determination itself when those requirements are followed to their terminus without stopping arbitrarily.

Therefore: determination and intelligence are functionally equivalent. Determination is intelligence viewed from the output. Intelligence is determination viewed from the source. They are not merely correlated. They are the same thing described from two directions.

Step 4 — Physics confirms the terminus. No derived system produces more than its source provides. Intelligence at the operational floor — confirmed at 61 million observations — requires constitutive intelligence at the source. The regress terminates at a ground that is determined without being derived.

You cannot produce more than what is provided. Intelligence at the operational floor requires intelligence at the source. This is not inference. It is the conservation principle applied to organizational capacity.

The Darwin Falsification Statement

Charles Darwin specified his own falsification condition in Origin of Species, Chapter 6:

'If it could be demonstrated that any complex organ existed which could not possibly have been formed by numerous, successive, slight modifications, my theory would absolutely break down.'

The Anchor-Single pair analysis satisfies this condition on Darwin's own terms. The gap between functional and disordered pair preference matrices is not a gradient of successive slight modifications. It is a categorical organizational difference:

Property	Data Result
Functional vs disordered gap	Categorical, not continuous. $A \rightarrow N: Z = -87$ functional, $Z = +2.33$ disordered.
Intelligence as incremental?	No. Zero boundary-setting capacity produces zero boundary-setting. The threshold cannot be crossed by accumulation.
Darwin's condition satisfied?	Yes. The organizational capacity demonstrated in functional sequences cannot be formed by successive slight modifications from disordered sequences. The data shows categorical reversal, not gradient.
Darwin's own standard	His theory would absolutely break down. The data satisfies

Property	Data Result
	his stated condition on his own terms.

This is not a claim against Darwin's account of adaptation within functional systems. Adaptation is real and observable. The falsification applies specifically to the claim that functional semantic intelligence — the capacity to set and maintain organizational boundaries — arises incrementally from systems that do not possess it.

Part II: Theorem — The Trinitarian Minimum

[Inseparability of M/I/C] In a universe where $\epsilon_{sys} > 0$ (the Second Law), a system cannot maintain $X_f > 0$ if any component of the M/I/C triad is absent.

2.1 Absence of Intelligence (Imin = 0)

The system possesses Valence (M) but no corrective mechanism. Environmental noise causes signal drift.

$\Delta I_{sys} \rightarrow 0 \implies X_f(t)$ dominated by $-\epsilon_{sys}$. Result: Preference exists but cannot be actualized. The system dissolves.

2.2 Absence of Valence (Mmin = 0)

The system possesses Predictive Correction (I) but no criterion for success. In a vast search space (Ω_{sys}), the system performs a random walk.

$\Phi \cdot \Delta I_{sys} / \Omega_{sys} \ll \epsilon_{sys} \implies X_f(t) \leq 0$. Result: Computation without goal-direction is indistinguishable from heat.

2.3 Absence of Loop Closure (Cmin = 0)

The system has M and I but lacks a unified standpoint. Without the Translation Protocol (Cmin), DNA and amino acids in the same solution will not interact symbolically. The ribosome is not merely a catalyst — it is the enforcement mechanism of loop closure.

2.4 The Logic Audit Summary

Step	Consequence
$\Phi = 0$ (pure stochasticity)	Random Walk fails due to Ω_{sys} and ϵ_{sys} . $X_f \rightarrow 0$.

Step	Consequence
$X_f > 0$ is observed (life exists)	Φ must be non-zero. This is not inferred — it is required.
$\Phi > 0$	M, I, and C are functionally present. The triad is the minimal decomposition of Φ .

Part III: The Causal Continuity Axiom — A Derived Conclusion

The theorem establishes that $\Phi > 0$ is required for any functional system. But a sharp objection arises: if Cmin is instantiated biologically by the ribosome, and the ribosome evolved, then the M/I/C triad was incomplete before the ribosome existed. What guided pre-ribosomal stages?

The Causal Continuity Axiom addresses this directly. It is presented here not as an axiom requiring acceptance but as a conclusion supported by five independent derivations. The reader is invited not to accept it as a premise but to identify which of the five derivations fails.

The Causal Continuity Axiom: Derived systems cannot exceed the causal capacity of their ground. If $\Phi > 0$ is required for any functional transition, and functional transitions are observed continuously across all stages of molecular evolution, then $\Phi > 0$ must be continuous across those stages — not episodic.

Five Independent Derivations Supporting the Axiom

First — The Determination Chain

Determination requires a comparator. A comparator requires discernment. Discernment requires a criterion maintained consistently across instances. A criterion maintained consistently is not episodic — it is continuous. Therefore determination, confirmed at 61 million Anchor-Single pair observations, requires continuous criterion-governed constraint. If Φ were episodic — present at some stages and absent at others — determination would be episodic. The data shows determination is not episodic. Therefore Φ is not episodic. Therefore the Causal Continuity Axiom follows from the determination chain established in Section 1.4.

Second — The Conservation Argument

The First Law of Thermodynamics requires a consistent criterion maintained across every interaction. Conservation is not episodic. It doesn't hold sometimes and fail at others. It holds continuously across every point in the system simultaneously. If the organizational capacity that conservation requires were episodic — absent at some stages — conservation would fail at those stages. Conservation doesn't fail. Therefore

organizational capacity is continuous. Therefore $\Phi > 0$ is continuous. The Causal Continuity Axiom follows from conservation itself.

Third — The Threshold Discontinuity Argument

The three thresholds (T_i , T_p , T_r) are discontinuities, not gradients. Below Eigen's error threshold, functional information does not accumulate slowly — it catastrophically degrades. A system cannot inch toward a discontinuous threshold while below it. The only coherent explanatory model for threshold crossing is one in which Φ was present prior to and through the crossing.

Fourth — Direct Measurement of Delta-C

EMPIRICAL ANCHOR: $\Delta C > 0$ at all 16 measurement points across four biological organisms and four complexity levels. No biological system has ever been observed operating at the maximum entropy baseline. The denial of the Axiom now requires positing an unobserved causal discontinuity in direct contradiction of 16 independent empirical measurements all showing the opposite pattern (Van Slyke, 2026b).

Fifth — The Vessel Principle

No derived system produces more than its source provides. If Φ were episodic — absent at inception and appearing later — then at the transition point the system would produce $\Phi > 0$ from a source with $\Phi = 0$. That requires the output to exceed the input. That violates conservation applied to organizational capacity. Therefore Φ cannot have transitioned from zero. Therefore it was always present. Therefore it is continuous.

3.1 The Upgrade: From Inference to Logical Necessity

If $\Phi = 0$ produces only dissolution, and $\Phi > 0$ is observed at every stage of functional history, then $\Phi > 0$ must have been causally present before the first functional stage. It cannot have transitioned from zero. Nothing emerges from zero.

3.2 The Stepwise Emergence Objection

The stepwise account faces a precise technical barrier: the threshold problem. Each of the three thresholds (T_i , T_p , T_r) has a discontinuous character. Partial function at sub-threshold levels does not persist long enough to be selected, because ϵ_{sys} continuously degrades it. The system cannot inch toward a threshold while below it — it dissolves before arriving.

Part IV: Formalization of Required Functionality

4.1 Threshold T_i — Specified Functional Information

$$I_f = -\log_2 (N_f / N_{total})$$

Where N_f is the number of sequences that perform the function and N_{total} is the total number of possible sequences. Axe (2004) estimated that for a domain-sized functional protein fold, $N_f / N_{total} \approx 10^{-77}$, giving $I_f \approx 256$ bits. Note: this estimate is contested in the molecular evolution literature, with critics noting it was derived from a single protein family and may not generalize. The temporal constraint argument in Section 4.6 provides an independent and less contestable basis for the search space problem that does not depend on this specific estimate.

4.2 Threshold T_p — Active Persistence ($\Phi > 0$)

$$\Phi \cdot \Delta I_{sys} / \Omega_{sys} > \epsilon_{sys}$$

Below this threshold, any functional information accumulated is degraded faster than it can be maintained. The T_p threshold separates passive chemical aggregates from actively persistent functional systems.

4.3 Threshold T_r — Error-Correcting Replication

$$\mu < 1/L \text{ (Eigen's error threshold)}$$

Above this threshold, mutational load exceeds the system's capacity to maintain functional sequence identity. Functional information degrades across generations rather than accumulates.

4.4 The Strong/Weak Functional Bias Distinction

Type of Bias	Requires Reference?	Requires Error Correction?	Entailed by Physics Alone?
Weak (law-driven): Crystal formation, lipid vesicles	No	No	Yes
Strong (reference-dependent): Genetic code, ribosomal translation	Yes	Yes	No

4.5 Abiogenesis Model Mapping

The three thresholds give critics a precise, measurable target: demonstrate a pathway from non-living chemistry to a system satisfying all three simultaneously, without invoking selection by criterion at any step.

Model	Ti (Spec. Info)	Tp (Active Φ)	Implicit Criterion?
RNA World	Partial (lab-selected)	Partial	Yes — catalytic feedback + persistence criterion
Metabolism First (Hydrothermal)	Not crossed	Partial (energy gradients)	Yes — thermodynamic viability selection
Lipid World / Compartmentalization	Not crossed	Partial	Yes — differential growth rate
Crystal / Mineral Surface	Not crossed	Not crossed	Yes — 'better scaffold' criterion
Panspermia / Seeding	Deferred	Deferred	No attempt — relocates the question

The pattern is uniform: every model that makes genuine progress invokes something functionally equivalent to a directing criterion at the precise point of threshold crossing. No model yet demonstrates all three thresholds satisfied without one. The burden on naturalistic models is precise and falsifiable: show the crossing without the criterion.

4.6 The Temporal Closure — Why Geological Time Does Not Solve the Search Space Problem

A persistent response to search space objections invokes geological time and large population sizes as sufficient to traverse the functional sequence space. This response fails on biological grounds that are prior to and independent of any probability estimate from the literature.

Individual organisms are temporally bounded systems. Variation must implement within the lifespan of the organism or it is lost entirely. Death does not extend the search. It resets it. This produces three precise consequences for the gradualist account.

Consequence 1 — No memory across resets. Populations do not accumulate progress toward functional thresholds across generations. Individual organisms do not carry forward the organizational memory of non-functional intermediate states. Each organism is an independent search with a fixed window. When the organism dies the search terminates. The next organism begins without memory of what the previous search attempted or approached. Geological time does not expand the search space

available to any individual organism. It multiplies the number of independent resets. The traversal problem is not solved by time. It is compounded by it.

Consequence 2 — Blind variation within every lifespan. Mutation, recombination, and copying errors have no discernment, no memory, and no foresight. They cannot target functional relevance within the available window. They cannot preferentially generate variations that approach a threshold. Each variation event is independent of previous variation events. The blind machinery does not learn. It generates variation without reference to the threshold it would need to cross.

Consequence 3 — Selection cannot act below the threshold. Natural selection requires functional variance to act on. Below the minimum viable function threshold — T_p in this framework — there is nothing to select. A variation that moves the organism toward a threshold but does not cross it within the lifespan confers no selectable advantage. It dies with the organism. The incremental improvement argument fails at the precise point it is needed most — below T_p , where selection cannot yet act and blind machinery operates without memory in a fixed window.

Geological time does not solve the search space problem because time is not the variable that limits the search. The variable that limits the search is the lifespan of the individual organism. Each organism is a temporally bounded independent search. Death resets the search without memory. Populations multiply the number of resets — they do not accumulate progress toward thresholds. This closure does not depend on Axe (2004) or any contested probability estimate. It depends only on two uncontested biological observations: organisms die, and variation-generating machinery is blind. Neither is disputed by anyone in the relevant literature.

Part V: Five Sealed Technical Closures

5A — The Frozen Accident Fallacy

The genetic code is not arbitrary. Comparative analyses (Freeland & Hurst, 1998) show the canonical code minimizes the impact of point mutations far more effectively than random alternative codes — placing it in approximately the top 1 in a million possible mappings for error minimization. For selection to refine a mapping into the near-optimal structure observed, the code itself would have to change while the organism is using it — which is usually fatal. The optimization was a prerequisite for functional persistence, not its downstream product.

5B — The Stereochemical Insufficiency

The observed affinities between codons and amino acids are weak, inconsistent, and present for only a small subset. The existence of tRNA adaptors demonstrates the mapping is not based on direct chemical attraction. Chemical affinity can produce preference. It cannot produce symbolic mapping. No accumulation of weak affinities can

generate a full translation system because the mapping must be complete and chemically meaningful before functional proteins can exist.

5C — Meaning Cannot Be Reduced to Function

Symbolic systems possess counterfactual sensitivity: the mapping would remain the same even if the chemistry changed. The codon AUG would still initiate translation even if the physical substrate were altered. This is the hallmark of a rule-governed system. Rules govern causal chains. They are not reducible to them. Function depends on meaning — therefore meaning cannot be reduced to function.

5D — Emergence Cannot Cross the Interpretive Boundary

Type	Status
Weak Emergence: System-level behavior fully entailed by properties of parts	Valid — but produces no new causal powers
Strong Emergence: New causal powers not reducible to the parts	Rejected by physicalism as a violation of causal closure
Interpretation: The capacity to treat one thing as representing another	Requires strong emergence — which physicalism cannot permit

5E — The Physics Objection Confirms Rather Than Escapes CI

A persistent objection holds that thermodynamic constraints and physical law explain biological constraint density without invoking CI. This objection does not escape the framework. It confirms it at a deeper level — through three independent sub-arguments.

Sub-argument E1 — Conservation requires discernment. The First Law requires a consistent criterion maintained across every interaction — distinguishing states that satisfy conservation from states that do not, without deviation. That is criterion-governed differential constraint. That is discernment by the functional definition derived in Section 1.4. The First Law does not merely fail to prohibit intelligence. It requires it at every point.

Sub-argument E2 — Mathematics is a fundamental comparator. Physical law is expressed mathematically. Mathematics is not a neutral descriptive overlay applied to pre-existing unintelligent reality. Mathematics is constituted by comparison at its irreducible core. Equality, inequality, symmetry, invariance — every mathematical operation requires that this state be evaluated against that state according to a criterion. A system that escapes the comparator by appealing to physical law escapes into mathematics. And mathematics is the comparator. Every term available to the objection — symmetry, equivalence, invariance, conservation — is constituted by comparison. To use symmetry to eliminate the comparator is to redefine what symmetry means.

Sub-argument E3 — The regress through physics terminates in the same place. Thermodynamic constraints presuppose mathematical structure. Mathematical structure presupposes stable logical relationships. Stable logical relationships presuppose that distinctions are possible and conserved. The conservation of distinctions — the condition that makes difference possible and persistent — is precisely what constitutive intelligence defines as its ground condition.

England's dissipation-driven adaptation (2013) is the strongest thermodynamic naturalist alternative. England's mechanism provides a physical basis for statistical self-organization tendencies. It does not supply a reference signal (Mmin), error correction (Imin), or symbolic loop closure (Cmin). Furthermore, England's framework presupposes physical law with the character required to permit dissipative self-organization — precisely the unexamined constitutive ground Sub-argument E3 identifies.

Part VI: The Empirical Anchor

6.1 The Optimized Genetic Code

The Harvard DNA Storage Study (Church et al., 2012) demonstrated that DNA is a symbolic medium: the relationship between the chemical bases and the information encoded is arbitrary in precisely the information-theoretic sense. In information theory, an arbitrary symbolic mapping requires a prior protocol.

- DNA is a symbolic medium (arbitrary mapping confirmed)
- The mapping is optimized far beyond chance (error minimization confirmed)
- The optimization predates the selection mechanism that was supposed to produce it (front-loading implied)
- The tRNA adaptor enforces the mapping symbolically, not chemically (Loop Closure Cmin confirmed at molecular level)

6.2 Organizational Efficiency in Functional Protein Domains

EMPIRICAL ANCHOR: NOI_suppression = 0.912 in functional domains versus 0.402 in non-functional regions. $Z = 11.68$, $p = 1.54 \times 10^{-31}$. Functional domains use 11.41 distinct amino acids per 20-position window versus 12.83 in random sequences — narrower vocabulary, higher functional precision. This is the signature that $\Phi > 0$ predicts: informational flux directed toward functional outcomes rather than distributed stochastically (Van Slyke, 2026b).

6.3 The Anchor-Single Determination Data

NEW EMPIRICAL ANCHOR: Anchor-Single pair analysis across 61,320,087 pairs confirms non-randomness at the operational floor. Specific pairs

systematically over-represented ($A \rightarrow L$: $Z = 239.55$, ratio = 1.3364) and under-represented ($A \rightarrow N$: functional $Z = -87.13$, disordered $Z = +2.33$, delta = 89 points). 25 anti-sequential bonds in functional vs 14 in disordered. Latent potential: 0.727 vs 0.532. The decision gate is specific to functional constraint — not a general property of biological sequences. What functional domains reject, disordered regions accept. This is categorical, not continuous (Van Slyke, 2026d). EXTENDED CONFIRMATION (Van Slyke, 2026f): Cohort recall analysis across 848,022 functional and 103,802 disordered sequences establishes co-selection memory at five simultaneous classification levels. CO class recall $1.4549 \times$ IID ($Z = +471$) on 821,862 independent positions. Sequential memory confirmed: lag-1 autocorrelation -0.2505 (functional) vs -0.1608 (disordered), significant at lags 1–4, decaying to noise by lag 5 — cycling at secondary structure periodicity (~ 30 residues). Directionality deviates from uniform null ($p \ll 0.01$) in all classes. Functional sequences achieve $79 \times$ stochastic efficiency in anchor-based recall. CLR-corrected coupling confirms functional channel independence (mean $|r|$ 0.1901) exceeds disordered (0.1943) after removing compositional artifacts.

6.4 The Perfect Hit Rate — The Inductive Record

Every known functional semantic system traces back to prior intelligence. Not most. Not the majority. Every one. The naturalist account requires at least one confirmed exception. The empirical record contains none. The hit rate is not high. It is perfect.

Part VII: The Three-Layer Firewall

To satisfy the core tension between formal proof and metaphysical inference, the conclusions of this paper are structured in three distinct layers with explicitly different epistemic statuses.

Layer	Claim / Status / Discipline
Layer 1 — The Theorem (Formal / Deductive)	$X_f > 0$ requires $\Phi > 0$. The M/I/C triad is the minimal decomposition of Φ in any closed-loop control system. Includes: 61M Anchor-Single pairs confirm non-randomness at operational floor. Delta-C > 0 at 16/16 measurements. NOI_suppression $Z = 11.68$. Determination-intelligence functional equivalence derived in Section 1.4. The formal first-order logical axiomatization of constitutive intelligence is given in Van Slyke (2026e). These are Layer 1 empirical and logical results — not dependent on Layer 2 or Layer 3 inferences for validity.
Layer 2 — The Corollary (Logical Derivation)	$\Phi > 0$ was continuous across all stages of biological history. Supported by five independent derivations: the determination

Layer	Claim / Status / Discipline
	chain, conservation, the threshold discontinuity argument, direct measurement of Delta-C, and the vessel principle. The Causal Continuity Axiom is presented as a derived conclusion, not an assumed premise.
Layer 3 — The Inference (Abductive / Best Explanation)	The non-cumulative ground of Φ — constitutive intelligence — is the precondition of all persistent systems. It is not a derived system within reality but the source of the rules reality operates by. Offered explicitly as abductive inference. The Causal Continuity Axiom, if accepted, elevates this to deductive.

7.1 The Two-Tier Distinction: Derived vs. Constitutive Intelligence

Tier	Definition
Tier 1 — Derived Intelligence	Functional information-processing systems that operate within existence. Contingent. Require prior organizational capacity. Subject to the GEE. Φ is measurable. No derived system exceeds its source.
Tier 2 — Constitutive Intelligence	The organizational capacity identical with existence itself. Not a system operating within reality but the precondition of reality being structured at all. It does not operate within a framework of rules. It is the source of the rules. It does not increment.

7.2 Why This Is Not Special Pleading

Special pleading would be: 'everything requires a cause except this one thing I need to be uncaused.' That is arbitrary exemption.

What this argument states is categorically different. Derived intelligence requires prior cause because it is contingent and operational — it functions within a pre-existing framework. Constitutive intelligence requires no prior cause because it is not contingent. It is not a thing inside reality that processes signals. It is the precondition of contingency itself.

The causal regress rule — 'every functional system requires prior organization' — applies specifically to systems operating within a framework. It does not apply to the framework itself, for the same reason that asking 'what is north of the North Pole' is not a gap in geography but a category error.

7.3 The Vessel Principle

Constitutive intelligence does not increment through deployment. Physics confirms: no derived system produces more than its source provides. The vessel limits. The source does not expand.

What we measure in functional protein domains — selectivity, efficiency, latent pathways, anti-sequential bonds — is not intelligence becoming more intelligent. It is intelligence being less constrained by the vessel through which it expresses. Disordered regions are poorer vessels. Functional domains are better vessels. The source is the same.

7.4 The Metaphysical Inference — Abductive Status

We offer this as an abductive inference — inference to the best explanation — not as a deductive consequence of the theorem alone. The Causal Continuity Axiom, if accepted as a premise, elevates the conclusion from abductive to deductive. The reader is invited to evaluate the axiom on its merits — or more precisely, to identify which of the five independent derivations supporting it fails.

7.5 Relationship to Panpsychism

The CI framework converges with panpsychism at its conclusion — mind is foundational not emergent — but diverges at its foundation in a way that is methodologically significant. Panpsychism grounds foundational mind in subjective experience, treating micro-experiential properties as fundamental and requiring an account of how they combine into unified consciousness. The CI framework grounds foundational intelligence in functional criterion-governed differential constraint — measurable organizational capacity confirmed at 61 million observations.

This distinction has three precise consequences. First, the CI framework does not require the hard problem of consciousness as a premise. It requires only that determination is non-random at the operational floor. The hard problem was formulated without empirical evidence that matter is determinately organized at that level. The CI framework provides that evidence and thereby changes what requires explanation. Second, the CI framework does not encounter the combination problem that centrally challenges panpsychism. Panpsychism generates the combination problem by fragmenting consciousness at the micro level and requiring combination upward. That combination model requires outputs to exceed inputs at every combinatorial step — which violates the vessel principle. The CI framework starts with unified constitutive intelligence at the ground and asks how vessels express it through increasing organizational capacity. Nothing is being combined. Third, the convergence between CI and panpsychism is confirmatory: two independent frameworks arriving at the same conclusion by different routes strengthens the abductive inference that mind is foundational.

Part VIII: Falsifiability Condition

This argument would be falsified by a demonstrated experimental pathway from non-living chemistry to a functional symbolic translation system — one in which sequences are mapped to outputs through an enforced symbolic relationship, arising without prior intelligent design of any component of that system. The operational criterion for a passing result is precise: (1) all components of the translation system must arise from an undirected chemical environment; (2) the researchers' role must be limited to observation and measurement — not selection, design of reaction conditions, or sequence specification; and (3) the system must autonomously cross all three thresholds ($I_f > T_i \wedge \Phi > \Phi_{min} \wedge I(X;Y) > Tr$) without a criterion being imported at any step. No such demonstration currently exists.

Part IX: Consolidated Key Takeaways

#	Claim and Foundation
1	The GEE requires $\Phi > 0$ for any $X_f > 0$. Cybernetic framework + formal theorem (Part II).
2	Φ is minimally decomposed into M (Reference Signal), I (Correction), C (Loop Closure). Control theory (Part I).
3	The genetic code is optimized, not arbitrary — front-loaded Φ is required. Freeland & Hurst 1998 (Part V-A).
4	Stereochemistry cannot produce symbolic mapping — tRNA adaptors confirm Cmin. Biochemistry (Part V-B).
5	Meaning cannot be reduced to function; function depends on meaning. Biosemiotics (Part V-C).
6	Emergence cannot cross the interpretive boundary within physicalism. Philosophy of mind (Part V-D).
7	$\Phi > 0$ must be continuous across all stages — Causal Continuity Axiom supported by five independent derivations, not an assumed premise (Part III).
8	The empirical hit rate is perfect: every functional semantic system traces to prior intelligence. Inductive record (Part VI).
9	The regress terminates in constitutive intelligence — non-cumulative, non-contingent. Two-tier distinction (Part VII).
10	The metaphysical inference is abductive; the Causal Continuity Axiom elevates it to deductive. Epistemic firewall (Part VII).
11	$\Delta C > 0$ at all 16 biological measurements — structure never absent at any complexity level. Van Slyke 2026b, Section 1.3.

#	Claim and Foundation
12	Phi_CI = 0.2080 — empirically derived source ceiling; no system reaches it. Van Slyke 2026b, Section 1.3.
13	NOI_suppression 0.912 functional vs 0.402 non-functional — $\Phi > 0$ directly observed in protein domains. $Z = 11.68$, $p = 1.54 \times 10^{-31}$.
14	The physics objection confirms CI rather than escaping it through three independent sub-arguments: conservation requires discernment, mathematics is a fundamental comparator, and the regress through physics terminates in the same constitutive ground. Section 5E.
15	61,320,087 Anchor-Single pairs confirm non-randomness at operational floor. Non-random = determined. Determined = intelligent by functional derivation — not dictionary definition. Four sub-steps: comparator, discernment, criterion, functional equivalence. Section 1.4.
16	Darwin's own falsification condition is satisfied. Intelligence is not produced incrementally from non-intelligence. Zero boundary-setting produces zero boundary-setting. Darwin (Origin of Species, Chapter 6).
17	The decision gate is specific to functional constraint. A → N: functional $Z = -87.13$, disordered $Z = +2.33$. Categorical difference, not gradient. 25 vs 14 anti-sequential bonds. Latent potential 0.727 vs 0.532. Extended confirmation: cohort recall analysis across 848,022 functional sequences establishes co-selection memory at five simultaneous classification levels (CO class: $1.4549 \times \text{IID}$, $Z = +471$, on 821,862 independent positions). Coupling analysis (CLR-corrected) confirms functional sequences maintain more independent channels than disordered (mean $ r $ 0.1901 vs 0.1943). Sequential memory: lag-1 autocorrelation -0.2505 functional vs -0.1608 disordered, cycling at secondary structure periodicity. Efficiency: functional sequences achieve $79 \times$ stochastic efficiency in anchor-based recall (Van Slyke, 2026f).
18	The temporal closure: geological time does not solve the search space problem. Individual organisms are temporally bounded. Death resets the search without memory. Blind variation machinery cannot target thresholds within a lifespan. Selection cannot act below T_p . Section 4.6. Protein viability sharpens this closure: individual proteins are viable for hours. The co-selection memory architecture measured across five simultaneous classification channels is therefore a prerequisite for first successful protein folding — not a product of iterative selection. A sequence either carries the architecture and folds, or does not and leaves no offspring. The architecture cannot be assembled by selecting among non-functional sequences because non-functional sequences produce no heritable continuation (Van Slyke, 2026f).
19	Mathematics is a fundamental comparator. Every mathematical operation is constituted by comparison at its irreducible core. Every objection that retreats to physical law or mathematical structure to escape the comparator retreats deeper into comparison. Section 5E.
20	The CI framework converges with panpsychism at the conclusion but diverges at the foundation. CI grounds intelligence in measurable organizational constraint, not

#	Claim and Foundation
	subjective experience. The combination problem does not apply to CI because the vessel principle replaces combination with expression. Section 7.5.

References

Van Slyke, P.B. (2026a). The Logic Circuit: On the Logical Insufficiency of Unguided Process as a Complete Explanatory Framework for the Origin of Functional Semantic Intelligence. Natural Organization Institute.

Van Slyke, P.B. (2026b). Constitutive Intelligence: Intelligence at Inception and the Accumulation of Mind. Natural Organization Institute. Van Slyke, P.B. (2026f). A Domain-Agnostic Framework for Detecting Structure in Sequential Constrained System Van Slyke, P.B. (2026c). The Generalized Existence Theorem. Version 8. Natural Organization Institute. [This document]s. Natural Organization Institute. Van Slyke, P.B. (2026g). Empirical Convergence on the Coordination Ratio Φ : How Cohort Recall, Coupling, and Directionality Independently Map Onto the GEE's Theoretical Structure. Natural Organization Institute.

Van Slyke, P.B. (2026h). Absent from Observation: Constitutive Intelligence as the Unmeasurable Ground of Measurable Structure. Natural Organization Institute. Van Slyke, P.B. (2026). Formal Proof of IID Incompatibility: Support Violation and Structural Constraints in Number Generation. Version 14. Natural Organization Institute Technical Report. Van Slyke, P.B. (2026). Natural Organization: A Universal 90/10 Principle of Structure and Mutation. Submitted to Entropy, MDPI.

Van Slyke, P.B. (2026d). Anchor-Single Pair Analysis: The Decision Gate in Functional Protein Domains. Natural Organization Institute.

Van Slyke, P.B. (2026e). Cosmic Intelligence Model: Formal Logical and Symbolic Representation. Natural Organization Institute.

Church, G.M., Gao, Y., & Kosuri, S. (2012). Next-generation digital information storage in DNA. *Science*, 337(6102), 1628.

Freeland, S.J. & Hurst, L.D. (1998). The genetic code is one in a million. *Journal of Molecular Evolution*, 47(3), 238–248.

Crick, F. (1968). The origin of the genetic code. *Journal of Molecular Biology*, 38(3), 367–379.

Koonin, E. (2007). The cosmological model of eternal inflation and the transition from chance to biological evolution. *Biology Direct*, 2:15.

Orgel, L. (2004). Prebiotic chemistry and the origin of the RNA world. *Critical Reviews in Biochemistry and Molecular Biology*, 39(2), 99–123.

Chalmers, D. (1995). Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2(3), 200–219.

Chalmers, D. (1996). *The Conscious Mind*. Oxford University Press.

- Nagel, T. (2012). *Mind and Cosmos*. Oxford University Press.
- Goff, P. (2019). *Galileo's Error*. Pantheon Books.
- Shannon, C. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*, 27, 379–423.
- Meyer, S. (2009). *Signature in the Cell*. HarperOne.
- Meyer, S. (2013). *Darwin's Doubt*. HarperOne.
- Darwin, C. (1859). *On the Origin of Species*. John Murray.
- Yockey, H. (2005). *Information Theory, Evolution, and the Origin of Life*. Cambridge University Press.
- Eigen, M. (1971). Self-organization of matter and the evolution of biological macromolecules. *Naturwissenschaften*, 58(10), 465–523.
- Axe, D.D. (2004). Estimating the prevalence of protein sequences adopting functional enzyme folds. *Journal of Molecular Biology*, 341(5), 1295–1315. [Note: probability estimates from this source are contested in the molecular evolution literature. Cited as supporting context; the temporal constraint argument in Section 4.6 provides an independent basis not dependent on this estimate.]
- Henikoff, S. & Henikoff, J.G. (1992). Amino acid substitution matrices from protein blocks. *PNAS*, 89(22), 10915–10919.
- Woese, C.R. (2002). On the evolution of cells. *PNAS*, 99(13), 8742–8747.
- Kim, J. (1999). Making sense of emergence. *Philosophical Studies*, 95(1–2), 3–36.
- Barbieri, M. (2003). *The Organic Codes*. Cambridge University Press.
- England, J.L. (2013). Statistical mechanics of self-replication. *Journal of Chemical Physics*, 139(12), 121923.
- Tononi, G. et al. (2016). Integrated information theory. *Nature Reviews Neuroscience*, 17(7), 450–461.
- Kauffman, S. (1995). *At Home in the Universe*. Oxford University Press.
- Schrödinger, E. (1944). *What Is Life?* Cambridge University Press.

Appendix A: Empirical Convergence on Φ — Summary Only (Full derivation, sensitivity table, and independent Anchor-Single confirmation in Van Slyke (2026g))

This appendix summarizes the empirical convergence analysis in Van Slyke (2026g). The formula $R \times C \times D$ does not substitute for Φ in the GEE. It demonstrates that three independently motivated measurements converge on the GEE's predicted structure.

A.1 Three Measurements, Three GEE Components

R (Recall Enhancement, CO class = 1.4549× IID) → Mmin: The system maintains a consistent criterion for which amino acid pairs appear together. That criterion is the reference signal.

C (Channel Independence, CLR mean $|r|$ = 0.1901 functional vs 0.1943 disordered) → Imin: Functional sequences maintain more orthogonal channels. More orthogonal = more precise simultaneous constraint = higher comparator efficiency.

D (Directionality Deviation, $\chi^2 = 307,168$ on 1.06×10^6 positions, lag-1 = -0.2505) → Cmin: The comparator is an actively maintained loop cycling at secondary structure periodicity — the feedback integration loop closure requires.

A.2 What the Convergence Confirms

All three factors non-zero in both datasets: $\Phi > 0$ confirmed. Causal Continuity Axiom empirically supported. Functional exceeds disordered in all three dimensions simultaneously: vessel principle confirmed. All Φ_{abs} estimates below $\Phi_{CI} = 0.2080$ across full Φ_{max} range (1.5–1.8): source always exceeds expression. The 79× stochastic efficiency is a downstream consequence: high R (more recall per anchor), high C (no channel interference), high D (criterion does not drift). See Van Slyke (2026g) for full derivation, sensitivity table, and independent Anchor-Single confirmation.

The GEE predicted a structure. Independent measurements found it. The convergence was not designed in.